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Pressure Mounting of Fragile Silk Embroideries: A Reversible Approach Using Profiled Rigid Supports

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ABSTRACT

Fragile silk embroideries with brittle grounds present a persistent challenge in exhibition mounting: the object requires stable support, but any pressure on the embroidery surface risks compressing the raised pile and distorting the metallic thread structure. This paper presents a method using profiled rigid supports contoured to the precise relief of the embroidery surface, distributing the object's weight without contact with raised elements. The supports are fabricated from close-cell polyethylene foam profiled with a heated tool, covered with Hollytex, and held in position by a cotton lacing system attached to the mounting board. No adhesive contacts the textile at any point. Twelve case studies are presented from Ottoman, Chinese, and European embroidery traditions. All objects were remounted or uninstalled within the study period, and removal in every case was achieved without tool assistance in under thirty minutes, meeting the reversibility criterion.

1. INTRODUCTION

Textile mounting has long occupied an uncomfortable middle ground between conservation and display. The standard practice of stitching a textile to a fabric-covered board is reversible in principle but often destructive in execution when the ground fabric is fragile or when the surface embroidery is raised and structurally at-risk from any directional pressure.

The Ottoman embroideries in European collections present a consistent subset of this challenge. Ground fabrics in these objects are often linen or cotton, sometimes degraded by iron-gall treatments or localized pest damage, while the silk embroidery itself is layered in multiple passes that create a raised, three-dimensional surface texture. Any mounting system that places the object face-down, or applies even light pressure from behind, risks mechanical abrasion against the raised elements.

2. METHOD

Profiled supports were fabricated on-site for each object using a heated awl and dental tool to form shallow channels in 25 mm polyethylene foam corresponding to the locations of the raised embroidery elements. The profile was mapped by placing the foam directly under the object face (protected by a Mylar barrier) and pressing gently to mark contact points, then removing material in those areas to a depth of 3 to 5 mm.

3. RESULTS

Of twelve objects mounted using this method over a thirty-month study period, eleven remained stable through the exhibition cycle. One object (Case 7, a Chinese Qing-dynasty panel) experienced minor weft displacement along a pre-existing repaired join, attributed to insufficient lateral support rather than the profiling method itself. That object was remounted with an extended lacing system and remained stable thereafter.

Removal time across all twelve objects ranged from 12 to 28 minutes. No tool assistance was required in any case. No adhesive residue was present on any object after removal. No embroidery elements showed mechanical compression or abrasion.

4. CONCLUSION

Profiled rigid support mounting provides a viable alternative to direct-face pressure mounts for raised silk embroideries. The method requires more preparation time per object than standard linen-pocket systems but substantially reduces the risk of surface damage to three-dimensional embroidery elements. It is recommended for objects where the embroidery relief exceeds 4 mm or where the ground fabric is too fragile to accept any lateral tension.

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